

Enrique Noriega-Atala

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Education

PhD	The University of Arizona , Information Science • Minor: Statistics	Jan 2020
MS	The University of Arizona , Computer Science	Jan 2014
BS	Monterrey Institute of Technology and Higher Education , Information Technologies	Jan 2010

Experience

Northeastern University , Associate Research Scientist, Khoury College of Computer Sciences	Feb 2026 – present 8 months
Researcher in the Gyori Lab at Northeastern University, focusing on natural language processing and biomedical informatics with an emphasis on information extraction from scientific literature. • Develop models and datasets for information extraction in the biomedical domain. • Mentor graduate students whose focus of research includes Natural Language Processing. • Engage and liaise on behalf of Gyori lab with external research groups from the lab's research projects.	
The University of Arizona , Computational Sciences Researcher IV, Computer Science Department	Oct 2022 – Jan 2026 3 years 4 months
• Consult as NLP/LLM domain matter expert in research engagements between Computer Science and various academic departments. • Co-develop curriculum and teach workshops for graduate students, including Exploring the LLM Frontier and AI Makerspace. • Lead and collaborate on research projects about information retrieval, extraction, document modeling, program synthesis, etc. • Write and publish research work on top-tier, peer-reviewed venues. • Collaborate with planning and writing extramural research grants and other funding mechanisms. • Participate in the organization of academic workshops. • Mentor graduate students.	
The University of Arizona , Postdoctoral Research Associate, Computer Science Department	Jan 2021 – Sept 2022 1 year 9 months
• Collaborated on the development of five research publications in top-tier peer-reviewed venues. • Led a collaborative effort to develop a multi-disciplinary visual analytics tool for the visualization of information extraction in the biomedical domain. • Contributed to open source NLP projects. • Presented research projects in multiple symposiums and conferences. • Collaborated on the development of two grant proposals for NSF and DARPA. • Collaborated and coordinated the organization of two research workshops, one institutional and the other co-hosted at COLING '22. • Mentored graduate students in the Department of Computer Science.	

CodaMetrix , Senior NLP Scientist • Built a taxonomy of body regions, organs, and tissues using medical ontologies such as UMLS, SNOMED, and MeSH. • Bootstrapped a dataset using proprietary EHR corpora. • Implemented, trained, and tested state-of-the-art neural architectures with PyTorch and TensorFlow. • Developed APIs for integration with the company's product. • Participated as a mentor in the summer internship program and contributed to documentation and internal training curricula for NLP and machine learning.	Feb 2020 – Oct 2020 9 months
The University of Arizona , Graduate Research Associate Worked within CLU lab and ML4AI on NLP and Machine Learning projects applied to health sciences, contributing regularly to REACH, a machine reading system for biomedical publications.	Jan 2013 – Jan 2020 7 years 1 month
The University of Arizona , Teaching Assistant • Collaborated as a teaching assistant for graduate and undergraduate courses in Computer Science and the School of Information. • Graded assignments, exams, and projects, held office hours, designed assignments, and directed laboratory sessions and guest lectures. • Courses served: Computer Organization (2013), Data Structures (2014), Information Retrieval and Web Search (2015, 2016), Introduction to Machine Learning (2017, 2019).	Aug 2013 – Dec 2019 6 years 5 months

Honors and Awards

UA Graduate & Professional Student's Council Travel Grant	Jan 2018
UA School of Information's Travel Award	Jan 2018
International Conference in Data Mining Student Award	Jan 2018
Galileo Circle Scholar, UA's College of Science	Jan 2014
Instituto Educativo Sonora-Arizona Scholarship	2013–2014
CONACyT Graduate Studies Scholarship	2012–2014
CENEVAL's Outstanding Performance Testimony on the EGEL Test	Jan 2010

Journal Publications

- **Extracting Inter-sentence Relations for Associating Biological Context with Events in Biomedical Text.** *Enrique Noriega-Atala*, Paul D. Hein, Shraddha S. Thumsi, Zechy Wong, Xia Wang, Sean M. Hendryx, Clayton T. Morrison. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*. (2019).
- **Large-scale Automated Machine Reading Discovers New Cancer Driving Mechanisms.** Marco A. Valenzuela-Escárcega, Özgün Babur, Gus Hahn-Powell, Dane Bell, Thomas Hicks, *Enrique Noriega-Atala*, Xia Wang, Mihai Surdeanu, Emek Demir, Clayton T. Morrison. *Database: The Journal of Biological Databases and Curation*. (2018).

Conference Publications

- **When and Where Did it Happen? An Encoder-Decoder Model to Identify Scenario Context.** Enrique Noriega-Atala, Robert Vacareanu, Salena Torres Ashton, Adarsh Pyarelal, Clayton T. Morrison, Mihai Surdeanu. *Findings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. (2024).
- **Active learning design choices for NER with transformers.** Robert Vacareanu, Enrique Noriega-Atala, Gus Hahn-Powell, Marco A. Valenzuela-Escárcega, Mihai Surdeanu. *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING)*. (2024).
- **Visualizing Interaction Networks and Evidence in Biomedical Corpora.** Enrique Noriega-Atala, Md Rahat-Uz-Zaman, Ruchika Bhat, Mladen Jergović, Stephen G. Kobourov, Janko Nikolich-Žugich. *Proceedings of the 16th IEEE Pacific Visualization Symposium (PacificVis)*. (2023).
- **A Human-machine Interface for Few-shot Rule Synthesis for Information Extraction.** Robert Vacareanu, George C. Barbosa, Enrique Noriega-Atala, Gus Hahn-Powell, Rebecca Sharp, Marco A. Valenzuela-Escárcega, Mihai Surdeanu. *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*. (2022).
- **Learning what to read: Focused machine reading.** Enrique Noriega-Atala, Marco A. Valenzuela-Escárcega, Clayton T. Morrison, Mihai Surdeanu. *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*. (2017).

Workshop Publications

- **Variable Extraction for Model Recovery in Scientific Literature.** Chunwei Liu, Enrique Noriega-Atala, Adarsh Pyarelal, Clayton T. Morrison, Mike Cafarella. *Proceedings of the 1st Workshop on AI and Scientific Discovery: Directions and Opportunities (ACL)*. (2025).
- **Neural-Guided Program Synthesis of Information Extraction Rules Using Self-Supervision.** Enrique Noriega-Atala, Robert Vacareanu, Gus Hahn-Powell, Marco A. Valenzuela-Escárcega. *Proceedings of the Workshop on Pattern-based Approaches to NLP in the Age of Deep Learning at COLING*. (2022).
- **Learning Open Domain Multi-hop Search Using Reinforcement Learning.** Enrique Noriega-Atala, Mihai Surdeanu, Clayton T. Morrison. *Proceedings of the Workshop on Structured and Unstructured Knowledge Integration (ACL)*. (2022).
- **Low Resource Causal Event Detection from Biomedical Literature.** Zhengzhong Liang, Enrique Noriega-Atala, Clayton Morrison, Mihai Surdeanu. *Proceedings of the 21st Workshop on Biomedical Language Processing at ACL*. (2022).
- **Neural Architectures for Biological Inter-Sentence Relation Extraction.** Enrique Noriega-Atala, Peter M. Lovett, Clayton T. Morrison, Mihai Surdeanu. *Proceedings of Scientific Document Understanding at AAAI*. (2022).
- **Understanding the Polarity of Events in the Biomedical Literature: Deep Learning vs. Linguistically-informed Methods.** Enrique Noriega-Atala, Zhengzhong Liang, John A. Bachman, Clayton T. Morrison, Mihai Surdeanu. *Proceedings of the Workshop on Extracting Structured Knowledge from Scientific Publications at NAACL-HLT*. (2019).
- **Inter-sentence Relation Extraction for Associating Biological Context with Events in Biomedical Texts.** Enrique Noriega-Atala, Paul D. Hein, Shraddha S. Thumsi, Zechy Wong, Xia Wang, Clayton T. Morrison. *Proceedings of the Sixth Workshop on Data Mining in Biomedical Informatics and Healthcare at ICDM*. (2018).
- **Focused Reading: Reinforcement Learning for What Documents to Read.** Enrique Noriega-Atala, Marco A. Valenzuela-Escárcega, Clayton T. Morrison, Mihai Surdeanu. *Proceedings of the Interactive Machine Learning and Semantic Information Retrieval Workshop at ICML*. (2017).
- **Large-scale Automated Reading with Reach Discovers New Cancer Driving Mechanisms.** Marco A. Valenzuela-Escárcega, Özgün Babur, Gus Hahn-Powell, Dane Bell, Thomas Hicks, Enrique Noriega-Atala, Xia Wang, Mihai Surdeanu, Emek Demir, Clayton T. Morrison. *Proceedings of the BioCreative VI Workshop (BioCreative6 2017)*. (2017).

Conference Presentations

- **When and Where Did it Happen? An Encoder-Decoder Model to Identify Scenario Context.** (2024). Miami, Florida. Workshop on the Future of Event Detection at EMNLP 2024
- **Visualizing Interaction Networks and Evidence in Biomedical Corpora.** (2023). Seoul, South Korea. The 16th IEEE Pacific Visualization Symposium (PacificVis)
- **Neural-Guided Program Synthesis of Information Extraction Rules Using Self-Supervision.** (2022). Gyeongju, Republic of Korea. Workshop on Pattern-based Approaches to NLP in the Age of Deep Learning at COLING
- **Open Domain Multi-hop Search Using Reinforcement Learning.** (2022). Seattle, Washington. Workshop on Structured and Unstructured Knowledge Integration at NAACL-HLT
- **Low Resource Causal Event Detection from Biomedical Literature.** (2022). Dublin, Ireland. 21st Workshop on Biomedical Language Processing at ACL
- **Neural Architectures for Biological Inter-Sentence Relation Extraction.** (2022). Vancouver, Canada. Scientific Document Understanding at AAAI
- **Understanding the Polarity of Events in the Biomedical Literature: Deep Learning vs. Linguistically-informed Methods.** (2019). Minneapolis, Minnesota. Workshop on Extracting Structured Knowledge from Scientific Publications at NAACL-HLT
- **Inter-sentence Relation Extraction for Associating Biological Context with Events in Biomedical Text.** (2018). Singapore. Sixth Workshop on Data Mining in Biomedical Informatics and Healthcare at ICDM
- **Grounding Gradable Adjectives through Crowd-sourcing.** (2018). Miyazaki, Japan. Language Resources and Evaluation Conference
- **Learning what to read: Focused machine reading.** (2017). Copenhagen, Denmark. NLP Applications track, EMNLP

Other Presentations

- **AZX Chat Bot: An LLM-driven Application Implementation to Support Resilience Against Climate Change.** (2023-11). The University of Arizona. Resilience Informatics Workshop
- **Visualizing Interaction Networks and Evidence in Biomedical Corpora.** (2023-11). The University of Arizona. Immune Monitoring Symposium
- **Machine Learning and NLP for Medical Research and Public Health.** (2023-05). The University of Arizona. PHIRE Side Chat
- **Accelerating Research through Large Scale, Interpretable Machine Reading.** (2022-10). The University of Arizona. MCB Joint Seminar
- **Accelerating Research through Large Scale, Interpretable Machine Reading.** (2022-09). The University of Arizona. Applied Data Science Speak & Greet Speaker Series
- **What is Machine Learning?** (2015). Hermosillo, Mexico. Guest lecture for Knowledge-based Systems in Organizations, Systems and Industrial Engineering Program at Monterrey Institute of Technology and Higher Education

Service

- **Organization Committee Member.** The Second Workshop in Pattern-based Approaches to NLP in the Age of Deep Learning (Pan-DL), EMNLP (2023); Pattern-based Approaches to NLP in the Age of Deep Learning (Pan-DL), COLING (2022).
- **Program Committee Member.** The AAAI-22 Workshop on Scientific Document Understanding, AAAI (2022); SUKI: Structured and Unstructured Knowledge Integration, NAACL-HLT (2022); TextGraphs: Graph-based Algorithms for Natural Language Processing (14, 15, and 16).
- **Student Volunteer.** (2019). Minneapolis, Minnesota. NAACL-HLT

Grants

- **SKEMA: Scientific Knowledge Extraction and Model Analysis.** (2022-07 – 2025-12). Co-PI, DARPA (Award amount: \$9,345,747)
- **Resilience Informatics for Public Health.** (2023-07 – 2024-06). Key Personnel, Technology and Research Initiative Fund (TRIF), The University of Arizona (Award amount: \$95,000)

Patents

- **Method for Extracting and Assessing Signaling Pathways from Biomedical Literature.** (2017-03). Provisional US patent filed. Authors: Mihai Surdeanu, Marco A. Valenzuela-Escarcega, Gustave Hahn-Powell, Dane Bell, Thomas Hicks, Enrique Noriega, Clayton Morrison